

# Fatemeh Fouladi Mahani

✦ **Montreal**, QC, Canada (Open to Relocation)

✦ Status: **Open Work Permit** (until Aug. 2026 and extendable)

✉ [fatemeh.fouladi.mahani@uottawa.ca](mailto:fatemeh.fouladi.mahani@uottawa.ca)

✉ [fatemeh.fouladi.eng@gmail.com](mailto:fatemeh.fouladi.eng@gmail.com)

 [linkedin.com/in/fatemeh-fouladi-mahani](https://www.linkedin.com/in/fatemeh-fouladi-mahani)

 [Google Scholar](#)

☎ +1-343-558-8746

## Summary

- **Ph.D. in Electrical Engineering** (November 2023) with **8+ years** of research experience in **nanophotonics**, **plasmonics**, and **integrated photonic device design**, with a strong focus on **metasurface engineering** and **photonic integration** for **optical sensing**, **biosensing**, **color filters**, and **telecommunications**
- Experience in **RF, optical, and THz communications**, including **antenna theory**, **electromagnetic modeling**, and the design of **photonic and RF front-end structures** for next-generation communication systems
- **Skilled researcher** with a track record of publishing **24+** papers in high-reputation journals and proceedings (**Cited 321+ times**, **h-index: 13**, and **i10-index: 16**)
- Demonstrated expertise in **experimental measurement** and **characterization** of **fabricated nanostructures** with a focus on **surface plasmon resonance (SPR) sensing** and **light-plasmon coupling**
- Proficient in theoretical design skills and **numerical simulations** with **Ansys Lumerical FDTD**, **COMSOL Multiphysics**, **Python**, and **MATLAB**, with additional experience in applying **Machine Learning** approaches to industrial projects
- **Outstanding academic performance** throughout B.Sc., M.Sc., and Ph.D. studies with top-tier grades, including a **19.50/20 in Ph.D.** and **19.43/20 in M.Sc.**, achieving **1<sup>st</sup> rank** in both degrees. Direct admission to M.Sc. and Ph.D. programs based on exceptional academic performance
- Recipient of multiple **prestigious awards** and **scholarships**, including the Scholarship Award for Talented Students, Outstanding Researcher Award, Visiting Research Scholarship Award, among others
- Strong background in **teaching**, **mentoring junior researchers**, and **leading interdisciplinary projects**; contributing to **10+ co-authored publications** on **light-trapping metasurfaces**, **graphene-integrated nanostructures**, and **photonic platforms** for **solar cells**, **sensors**, and **THz antennas**
- Experienced in writing and contributing to **research proposals** and engaging in **interdisciplinary collaborations**, with additional experience in **industry-driven research projects**

## Experience



### Research Assistant – International Collaboration

[Centre for Research in Photonics \(CRPuO\)](#), University of Ottawa

Feb. 2023 – Feb. 2024, Supervisor: [Prof. Pierre Berini](#)

- Led **optical characterization** and **experimental measurement** of **plasmonic gratings** for efficient **light-plasmon coupling** and **sensing**
- Designed **efficient plasmonic biosensors** on the end-facet of optical fibers for practical applications
- Simulated plasmonic pentamer-arranged nanohole arrays for **surface enhanced Raman scattering (SERS)**
- Published the outcomes in top-tier journals, including **Nature Scientific Reports**, **Physica Scripta**, and **Biosensors**
- Received three **scholarship awards** for this visiting program from the home country, Iran:
  - **Visiting Research Scholarship Award** – Ministry of Science, Research and Technology of Iran
  - **Foreign Research Opportunity Scholarship Award** – Iran's National Elites Foundation
  - **Visiting Student Support Scholarship** – Federation of Academic Pioneers in Iran



## Research Assistant

Shahid Bahonar University of Kerman and University of Ottawa

Sep. 2018 – Feb. 2023, Supervisors: *Prof. Pierre Berini* and *Prof. Arash Mokhtari*

- Designed and developed efficient plasmonic and photonic biosensors on the end-facet of dual-core optical fibers for practical applications

Shahid Bahonar University of Kerman

Sep 2015 - Sep 2018, Supervisors: *Prof. Arash Mokhtari* and *Prof. Mahdiyeh Mehran*

- Designed periodic structures for efficient plasmonic sensors, solar cells, and color filters
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## Industry-Based Research Programs

Selected through a competitive process for the 5<sup>th</sup> and 6<sup>th</sup> terms of Iran's National Elites Foundation Research Programs, focused on the application of **artificial intelligence (AI) / machine learning (ML)** approaches to address industry challenges

Nov. 2020 – Aug. 2021

- Developed an intelligent control system **integrating AI/ML** for the automated and continuous monitoring of hydrocyclone performance in mineral processing lines

Dec. 2021– Sep. 2022

- Achieved optimal design of semi-autogenous grinding (SAG) mill liners using **AI/ML**, focusing on wear pattern prediction of liners in the mineral processing line of Miduk Copper Mine (recognized among the top 3 research outcomes of the program)
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## Teaching Assistant

Shahid Bahonar University of Kerman, Department of Electrical Engineering

- Antenna Theory • Nanophotonics and Metamaterials • Microwave Engineering • Electromagnetics • Optical Communications
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## Skills

### ★ Core Expertise

#### Photonic Devices & Nanostructures

- Plasmonics • Nanophotonics • Integrated Photonics • Metasurfaces • Nanostructures • Photonic Device Engineering • Optical Fibers • 2D Materials • Quantum Light Sources

#### Sensing & Functional Platforms

- Optical Sensors • Biosensors • Lab-on-Fiber Platforms • Color Filters • Nanostructured Electrodes • Solar Cells • Spectral Filtering

#### Communications & Electromagnetics

- Optical Communications • RF Front-End Design • THz Wave Propagation • Antenna Theory • Electromagnetic Modeling

### ✂ Simulation & Design Tools

- Optical Simulation & Design • COMSOL Multiphysics • Ansys Lumerical FDTD / MODE • MATLAB • Python (NumPy, SciPy, Pandas, Matplotlib, scikit-learn) • Machine Learning • Signal Processing • Numerical Optimization Techniques

### 🔧 Experimental Techniques

- Experimental Measurement & Characterization of Nanostructures • Cleanroom Familiarity • Sample Preparation and Cleaning • Optical Setup Design & Alignment

## **Publications (Citations: 321+, h-index: 13, i10-index: 16), [Google Scholar](#)**

- [1] **F. F. Mahani**, L. A. M. Astorga, H. W. Choi, A. Mokhtari, and P. Berini, "Plasmonic slanted slit gratings for efficient through-substrate light-plasmon coupling and sensing," *Scientific Reports – Nature*, vol. 14, p. 2084, 2024.
- [2] **F. F. Mahani**, A. Mokhtari, and P. Berini, "Plasmonic biosensor on the end-facet of a dual-core single-mode optical fiber," *Biosensors*, vol. 13, p. 558, 2023.
- [3] **F. F. Mahani**, A. Mokhtari, and P. Berini, "Hybrid Si-Au plasmonic sensor on the end-facet of a dual-core optical fiber enhanced by hotspots: a theoretical study," *Physica Scripta*, vol. 99, p. 085523, 2024.
- [4] **F. F. Mahani**, M. Maleki, A. Mokhtari, and P. Berini, "Design of an efficient Fabry-Perot biosensor using high-contrast slanted grating couplers on a dual-core single-mode optical fiber tip," *IEEE Sensors Journal*, vol. 21, pp. 19705-19713, 2021.
- [5] Z. Khezripour, **F. F. Mahani**, and A. Mokhtari, "Performance improvement of thin-film silicon solar cells using transversal and longitudinal titanium nitride plasmonic nanogratings," *Optical Materials*, vol. 99, p. 109532, 2020.
- [6] M. Salemizadeh, **F. F. Mahani**, and A. Mokhtari, "Tunable mid-infrared graphene-titanium nitride plasmonic absorber for chemical sensing applications," *JOSA B*, vol. 36, pp. 2863-2870, 2019.
- [7] W. R. Wong, H. W. Choi, **F. F. Mahani**, L. A. M. Astorga, A. Mokhtari, and P. Berini, "Plasmonic pentamer-arranged nanohole arrays," in *2024 Photonics North (PN)*, 2024, pp. 1-2.
- [8] N. Kavooosi, M. Safinejad, M. Mehran, A. Mokhtari, and **F. F. Mahani**, "Design and fabrication of enhanced anti-reflective properties using pyramid/nanowire texturization of the silicon surface," *International Journal of Nanoscience and Nanotechnology*, vol. 18, pp. 241-250, 2022.
- [9] **F. F. Mahani** and A. Mokhtari, "Investigating the usage of nanohole array-based plasmonic silver electrodes in ITO-free organic solar cells," *Nanoscale*, vol. 6, pp. 26-32, 2019.
- [10] A. H. Kazemi, **F. F. Mahani**, and A. Mokhtari, "Peak amplitude enhancement of photoconductive antenna using periodic nanoslit and graphene in the THz band," *Optik*, vol. 185, pp. 114-120, 2019.
- [11] M. Salemizadeh, **F. F. Mahani**, and A. Mokhtari, "Design and development of efficient plasmonic sensors based on bi-layer of silver-SiO<sub>2</sub> nanodisk arrays," in *2019 27<sup>th</sup> Iranian Conference on Electrical Engineering (ICEE)*, 2019, pp. 311-314.
- [12] **F. F. Mahani**, A. Mahanipour, and A. Mokhtari, "Optimized design of nanohole array-based plasmonic color filters integrating genetic algorithm with FDTD solutions," *Journal of AI and Data Mining*, vol. 7, pp. 279-286, 2019.
- [13] M. Salemizadeh, **F. F. Mahani**, and A. Mokhtari, "Design of aluminum-based nanoring arrays for realizing efficient plasmonic sensors," *JOSA B*, vol. 36, pp. 786-793, 2019.
- [14] **F. F. Mahani** and A. Mokhtari, "Performance improvement of organic solar cells using a hybrid color filter electrode of graphene-aluminum nanorings," *Journal of Nanoelectronics and Optoelectronics*, vol. 13, pp. 1917-1923, 2018.
- [15] Z. Khezripour, **F. F. Mahani**, and A. Mokhtari, "Performance improvement of ultrathin organic solar cells utilizing light-trapping aluminum-titanium nitride nanosquare arrays," *Optical Materials*, vol. 84, pp. 651-657, 2018.
- [16] **F. F. Mahani** and A. Mokhtari, "TiO<sub>2</sub> circular nano-gratings as anti-reflective coatings and potential color filters for efficient organic solar cells," *Journal of Nanoelectronics and Optoelectronics*, vol. 13, pp. 1624-1629, 2018.
- [17] **F. F. Mahani** and A. Mokhtari, "Polarization-tuned chromatic electrodes using hybrid design of graphene-aluminum nanocross arrays for efficient organic solar cells," *Optical Materials*, vol. 84, pp. 158-165, 2018.
- [18] Z. Khezripour, **F. F. Mahani**, and A. Mokhtari, "Double-sided TiO<sub>2</sub> nano-gratings for broadband performance enhancement of organic solar cells," *JOSA B*, vol. 35, pp. 2478-2483, 2018.
- [19] **F. F. Mahani**, A. Mokhtari, and M. Mehran, "Design and development of aluminum nanoring arrays for realization of dual-mode operation plasmonic color filters," *JOSA B*, vol. 35, pp. 1764-1771, 2018.
- [20] Z. Khezripour, **F. F. Mahani**, and A. Mokhtari, "Optimized design of silicon-based moth eye nanostructures for thin film solar cells," in *2018 3<sup>rd</sup> Conference on Swarm Intelligence and Evolutionary Computation (CSIEC)*, 2018, pp. 1-4.
- [21] **F. F. Mahani** and A. Mokhtari, "Enhancement of ITO-free organic solar cells utilizing plasmonic nanohole electrodes," in *7<sup>th</sup> International Conference on Nanotechnology (ICN)*, 2017.

- [22] **F. F. Mahani**, A. Mokhtari, and M. Mehran, "Dual mode operation, highly selective nanohole array-based plasmonic colour filters," *Nanotechnology*, vol. 28, p. 385203, 2017.
- [23] **F. F. Mahani**, A. Mahanipour, and A. Mokhtari, "Optimization of plasmonic color filters for CMOS image sensors by genetic algorithm," in 2017 2<sup>nd</sup> Conference on Swarm Intelligence and Evolutionary Computation (CSIEC), 2017, pp. 12-15.
- [24] **F. F. Mahani** and A. Mokhtari, "Performance enhancement of nanohole array-based plasmonic color filters for CMOS image sensors," in 23<sup>rd</sup> Iranian Conference on Optics and Photonics (ICOP), and the 9<sup>th</sup> Iranian Conference on Photonics Engineering and Technology (ICPET), 2017.
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## Education



### **Shahid Bahonar University of Kerman**

Ph.D. in Electrical Engineering – Field and Wave Telecommunications

Sep. 2018 – Nov. 2023, Grade: 19.50/20.00, Thesis Score: 20/20, Supervisors: *Prof. Pierre Berini* and *Prof. Arash Mokhtari*



### **Shahid Bahonar University of Kerman**

M.Sc. in Electrical Engineering - Field and Wave Telecommunications

Sep. 2015 – Sep. 2017, Grade: 19.43/20.00, Thesis Score: 20/20, Supervisor: *Prof. Arash Mokhtari*, Advisor: *Prof. Mahdijeh Mehran*



### **Shahid Bahonar University of Kerman**

B.Sc. in Electrical Engineering - Telecommunications

Sep. 2011 – Sep. 2015, Grade: 16.16/20.00

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## Selected Honors & Awards

- **Scholarship Award for Talented Students** – Iran's National Elites Foundation (Dec. 2018 – Feb. 2022)
  - **Excellence in Research Award** – Kerman Province (Dec. 2018)
  - **Foreign Research Opportunity Scholarship Award** – Iran's National Elites Foundation (Feb. 2023 – Aug. 2023)
  - **Visiting Research Scholarship Award** – Ministry of Science, Research and Technology of Iran (Feb. 2023 – Aug. 2023)
  - **Outstanding Researcher Award** – Shahid Bahonar University of Kerman (Dec. 2018)
  - **Direct M.Sc. and Ph.D. Admission without Examination** – Exceptional Talents Foundation, Shahid Bahonar University of Kerman
  - **First-Ranked M.Sc. and Ph.D. Student** – Shahid Bahonar University of Kerman
  - **Selected for the 5<sup>th</sup> and 6<sup>th</sup> terms of Iran's National Elites Foundation Research Programs** (Nov. 2020 – Sep. 2022)
  - **Visiting Student Support Scholarship** – Federation of Academic Pioneers in Iran (Feb. 2023 – Oct. 2023)
  - **Research Scholarship Offer** – from the University of Ottawa (Jun. 2023 – Jan. 2024)
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## Language Fluency

• English (Fluent/Advanced) • Farsi (Native) • French (Basic/Elementary)

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## Peer Review Contributions (Technical Reviewer)

Served as a **technical reviewer** for several prestigious journals, including:

• *Optics Express* • *JOSA A* • *Optics Letters* • *Optics Continuum* • *Optical Materials Express* • *JOSA B*